

Module 4 Lesson 2

Percent, Sales Tax, and Discounts (8.1)

Objectives

In this lesson, we will learn how to

- express a fraction as a percent;
- express a decimal as a percent;
- express a percent as a decimal;
- solve applied problems involving sales tax and discounts;
- determine percent increase or percent decrease;
- investigate some ways percent can be abused.

1. Basics of Percent

Percents are numbers expressed as parts of 100.

The word **percent** means **per hundred**.

$$45\% = \frac{45}{100}$$

Meaning of Percent

A percent compares a number to 100.

$$p\% = \frac{p}{100}$$

2. Expressing a Fraction as a Percent

Fraction to Percent

To express a fraction as a percent:

1. Divide the numerator by the denominator.
2. Multiply the quotient by 100. This moves the decimal point two places to the right.
3. Add a percent sign.

Example 1: Express

$$\frac{5}{8}$$

as a percent.

Solution

$$5 \div 8 = 0.625$$

$$0.625 \times 100 = 62.5$$

Thus,

$$\frac{5}{8} = 62.5\%.$$

3. Expressing a Decimal as a Percent

Decimal to Percent

To express a decimal as a percent:

1. Move the decimal point two places to the right.
2. Attach a percent sign.

Example 2: Express 0.47 as a percent.

Solution

$$0.47 = 47\%.$$

4. Expressing a Percent as a Decimal

Percent to Decimal

To express a percent as a decimal:

1. Move the decimal point two places to the left.
2. Remove the percent sign.

Example 3: Express each percent as a decimal:

$$19\%, \quad 180\%.$$

Solution

$$19\% = 0.19$$

$$180\% = 1.80 = 1.8$$

5. Percent Formula

Many applications involving percent are based on the following formula:

Percent Formula

$$\text{Amount} = \text{Percent} \times \text{Base}$$

In many word problems, the word **of** means multiplication.

For example,

$$20\% \text{ of } 80 = 0.20 \times 80.$$

6. Sales Tax

Sales tax is collected by states, counties, and cities on items sold to customers.

Sales Tax Formula

$$\text{Sales tax amount} = \text{tax rate} \times \text{item's cost}$$

Example 4: Suppose the local sales tax rate is 7.5%, and you purchase a bicycle for \$894.

- (a) How much tax is paid?
- (b) What is the bicycle's total cost?

Solution

$$7.5\% = 0.075$$

$$\text{Sales tax amount} = 0.075 \times 894 = 67.05$$

The tax paid is

$$\$67.05.$$

The total cost is

$$\$894.00 + \$67.05 = \$961.05.$$

Therefore, the bicycle's total cost is

$$\$961.05.$$

7. Discounts and Sale Price

Businesses often reduce prices, or give discounts, to attract customers and reduce inventory.

The **discount rate** is a percent of the original price.

Discount Formula

$$\text{Discount amount} = \text{discount rate} \times \text{original price}$$

Sale Price Formula

$$\text{Sale price} = \text{original price} - \text{discount amount}$$

Example 5: A gaming laptop with an original price of \$1460 is on sale at 15% off.

- (a) What is the discount amount?
- (b) What is the laptop's sale price?

Solution

$$15\% = 0.15$$

$$\text{Discount amount} = 0.15 \times 1460 = 219$$

The discount amount is

$$\$219.$$

The sale price is

$$\$1460 - \$219 = \$1241.$$

Therefore, the gaming laptop's sale price is

$$\$1241.$$

8. Percent Increase and Percent Decrease

When a quantity changes, we can describe the change using percent increase or percent decrease.

Percent Increase

$$\text{Percent increase} = \frac{\text{new amount} - \text{original amount}}{\text{original amount}} \times 100\%.$$

Percent Decrease

$$\text{Percent decrease} = \frac{\text{original amount} - \text{new amount}}{\text{original amount}} \times 100\%.$$

Example 6: In 2000, world population was approximately 6 billion.
Using projection data:

- (a) Find the percent increase in world population from 2000 to 2150 using the high-projection data.
- (b) Find the percent decrease in world population from 2000 to 2150 using the low-projection data.

9. Example of Percent Increase

Suppose the high projection for world population in 2150 is 30 billion.
The original amount was 6 billion.

$$\begin{aligned}\text{Percent increase} &= \frac{30 - 6}{6} \times 100\% \\ &= \frac{24}{6} \times 100\%\end{aligned}$$

$$= 4 \times 100\%$$

$$= 400\%.$$

The projected percent increase is

$$400\%.$$

10. Example of Percent Decrease

Suppose the low projection for world population in 2150 is 4 billion.

The original amount was 6 billion.

$$\text{Percent decrease} = \frac{6 - 4}{6} \times 100\%$$

$$= \frac{2}{6} \times 100\%$$

$$= \frac{1}{3} \times 100\%$$

$$= 33\frac{1}{3}\%.$$

The projected percent decrease is

$$33\frac{1}{3}\%.$$

11. Abuses of Percent

Percent can sometimes be misleading if it is not used carefully.

One common mistake is confusing a difference in percentage points with a percent increase.

Percent Increase vs. Percentage Points

If a value changes from 4% to 13%, the increase is

$$13\% - 4\% = 9$$

percentage points.

But the percent increase is

$$\frac{13 - 4}{4} \times 100\% = 225\%.$$

Example 7: John Tesh, while co-anchoring Entertainment Tonight, reported that the PBS series The Civil War had an audience of 13%, compared with the usual 4% PBS audience. He described this as “an increase of more than 300%.”

Did he report the percent increase correctly?

Solution

$$\text{Percent increase} = \frac{13 - 4}{4} \times 100\%$$

$$= \frac{9}{4} \times 100\%$$

$$= 2.25 \times 100\%$$

$$= 225\%.$$

The percent increase was 225%, not more than 300%. Therefore, the report was not correct.

Summary

Key Ideas

- Percent means per hundred.
- To convert a fraction to a percent, divide and multiply by 100.
- To convert a decimal to a percent, move the decimal point two places to the right.
- To convert a percent to a decimal, move the decimal point two places to the left.
- Sales tax is found by multiplying the tax rate by the item's cost.
- A discount is found by multiplying the discount rate by the original price.
- Percent increase and percent decrease compare the amount of change to the original amount.
- Percent can be misleading when percentage points are confused with percent increase.